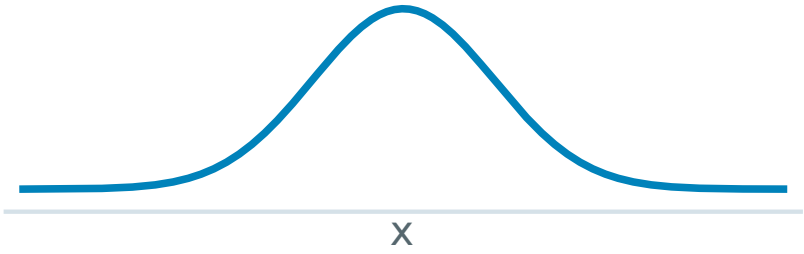
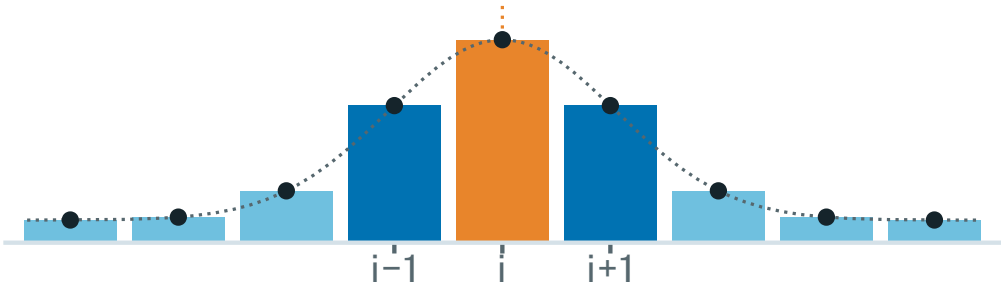
1 Continuous field $u(t_k, x)$



2 Discretise: grid values u_i^k



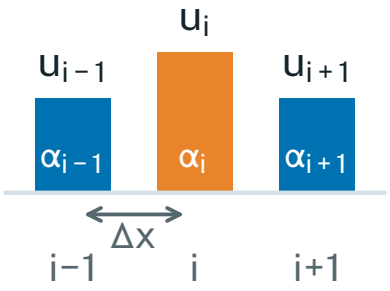
3 Stencil at cell i

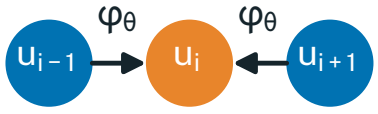
$$f_i = \frac{1}{\Delta x^2} \sum_j \alpha_j u_j$$

classical: α_j fixed

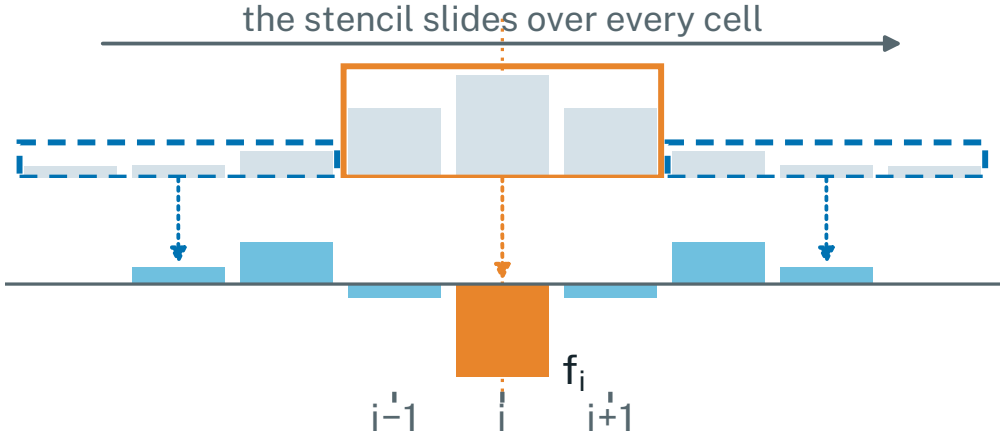
$$f_i = \psi_\theta\left(u_i, \sum_j \varphi_\theta(u_i, u_j)\right)$$

GNN: φ_θ, ψ_θ learned



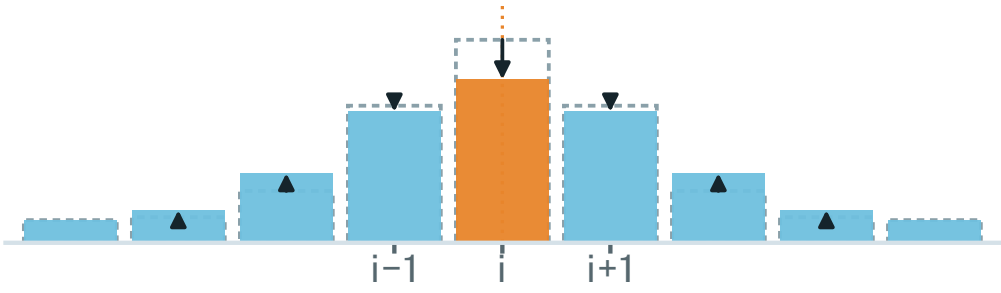


4 Slide the stencil: rates f_i



5 Time step

$$u_i^{k+1} = u_i^k + \Delta t\, f_i$$

explicit Euler; RK4 or any scheme that evaluates f can replace it

6 Next state u_i^{k+1} (dashed: u_i^k)